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Michigan Air Temperature by Population and Land Cover

Introduction

This project examines Michigan air temperature data and its relationship with population and land cover. In addition to better understanding how Michigan air temperature has changed over time (including years between 1970 and 2025) and how it differs by county and peninsula, exploring the relationship between air temperature and population and land cover can help make sense of the factors that might impact air temperature. Better understanding how human population, developed land, vegetation, and water might affect air temperature is important because air temperature affects so many environmental processes and is a crucial part of examining global warming.

Data

The air temperature data came from the National Oceanic and Atmospheric Administration air temperature dataset. The data covered a total of 1,037 stations across the state of Michigan, though not all stations had complete data that was used in this project. This data was collected for a total of 10 years – 1970, 1980, 1990, 1995, 2000, 2005, 2010, 2015, 2020, and 2025. Air temperature data was collected as a monthly summary at each station for each year. The main variables used were the station ID, station location name, station latitude and longitude, date, and average monthly temperature, which was “computed by adding the unrounded monthly maximum and minimum temperatures and dividing by 2. Given in Celsius or Fahrenheit depending on user specification. Missing if more than 5 days within the month are missing or flagged or if more than 3 consecutive values within the month are missing or flagged.” Air temperature was downloaded and used in Celsius, except for the first half of 1970 which was downloaded in Fahrenheit and then converted to Celsius.

After downloading the air temperature data in 6 month increments (due to size limits) and then compiling it all into one sheet, the variable names were made lowercase and the date column was separated into a month column and a year column. Rows that were missing the average monthly temperature value were removed, which decreased the number of observations from 29,679 to 17,349. Stations with fewer than 36 months of temperature observations were removed, which left 15,416 observations and decreased the number of stations from 316 (after removing missing temperature values) to 197. In order to match the stations to counties, the station names, latitudes, and longitudes were given to ChatGPT, and ChatGPT provided a list of the counties that corresponded with each station. The stations covered 79 of Michigan’s 83 counties. The counties were then matched to the upper or lower peninsula by consulting a map of Michigan’s counties.

The population data was compiled from 3 different sources from the U.S. Census Bureau and Michigan Statistical Abstract. Multiple sources were used to compile data for more years. It was broken down by county, and data for the years 1970, 1980, 1990, 1995, 2000, 2005, 2010, 2015, 2020, and 2025 was compiled. Besides compiling data from the different sources and choosing the years to use, no other cleaning was needed.

The land cover data came from the 2020 Land Cover (CONUS) file from the National Land Cover Database. After extracting data for Michigan counties, a dataset of land cover proportion by county and NLCD cover class was obtained. The data was originally broken down into 15 different cover classes, but some of these were combined to create a total of 6 class categories – water (class 11), developed land (classes 21-24), barren land (class 31), forest (classes 41-43), vegetation (classes 52, 71, 81-82), and wetlands (classes 90 and 95).

Methods

An ANOVA was run to compare the air temperatures across the state for the 10 different years. This technique was appropriate because 3 or more means were being compared. Air temperature was the variable being measured, and year was the factor whose effects were being investigated. For this test, the null hypothesis is that all 10 group means are not significantly different from each other, and the alternative hypothesis is that at least one group mean is significantly different from the rest. The purpose of this test was to see if air temperature changes significantly across different years.

Another ANOVA was run to compare air temperatures across cover class categories for 2020. This was another situation in which 3 or more means were being compared. Each county was labeled with the cover class category that made up the highest proportion of area in that county. This was the factor whose effects were being investigated, and air temperature was the variable being measured again. For this test, the null hypothesis is that all 6 group means are not significantly different from each other, and the alternative hypothesis is that at least one group mean is significantly different from the rest. The purpose of this test was to see if air temperature differs significantly depending on the primary land cover in the county.

A Kruskal-Wallis test was used to compare air temperatures across the 70 different counties included for 2025. This compares medians across 3 or more groups, and it is the nonparametric equivalent of an ANOVA. This test made more sense than an ANOVA since so many different groups were being compared. Air temperature was the variable being measured, and county was the factor whose effects were being investigated. The null hypothesis is that all 70 group medians are not significantly different from each other, and the alternative hypothesis is that at least one group median is significantly different from the rest. The purpose of this test was to see if air temperature differs significantly across different counties in the state.

Similarly, a t-test was run to compare air temperatures between the upper peninsula and lower peninsula. This test made sense because 2 different groups were being compared, with air temperature being the variable and peninsula being the factor. The null hypothesis is that the 2

group means are not significantly different, and the alternative hypothesis is that the 2 group means are significantly different from each other. This test shows whether air temperature in the upper peninsula is significantly different from air temperature in the lower peninsula.

A regression was run to examine the relationship between air temperature and population. This technique made sense because it explores the association between two continuous variables. Annual air temperature values and annual population values by county for the 10 different years were included. This regression was also run with log transformed population values, as this gave the residuals vs fitted, scale-location, and residuals vs leverage diagnostic plots a more even spread. A scatterplot with a fitted linear trend line was created to visualize this relationship.

More regressions were run to examine the relationship between air temperature and proportion of land covered for each of the 6 land cover class categories. These regressions again explored the association between two continuous variables. Using data from 2020, each county's annual average air temperature value and proportion of land covered by that cover class category were included in the regression. These regressions were also run with log transformed cover class category proportions to see which model produced a better fit. Scatterplots with fitted linear trend lines were created to visualize these relationships.

A correlation was run for air temperature and population values across all counties and all 10 years. This was used to examine the strength of the relationship between these two continuous variables. Correlations were also run for air temperature and each of the 6 land cover class categories to examine the strength of the relationship between air temperature and proportion of land covered by that category for 2020. Since the land cover data only corresponded to 2020, air temperature data for only 2020 was included.

Results

The ANOVA for air temperature across years yielded a p-value of 0.0012, indicating that at least one group mean is significantly different from the rest. Examining Tukey HSD indicated that the differences between 1980 and 1990, 1980 and 2010, and 1980 and 2020 were statistically significant, with the 1980 group mean being lower in each of these pairs.

The ANOVA for air temperature across cover class categories had a p-value of 0.0002, indicating that at least one group mean is significantly different from the rest. Tukey HSD indicated that the differences between vegetation and forest and vegetation and wetlands were statistically significant, with the vegetation group mean being higher in both of these pairs.

The Kruskal-Wallis test for air temperature across counties yielded a p-value of 0.9999, indicating that none of the group medians are significantly different from the rest. The air temperature medians for the counties are not significantly different from each other.

The t-test for air temperature between peninsulas gave a p-value of 2.2×10^{-16} , indicating that there is a statistically significant difference between the mean air temperature values for the upper and lower peninsula. The mean for the lower peninsula is 8.28°C , and the mean for the upper peninsula is 5.36°C . The 95 percent confidence interval is $[2.38^{\circ}\text{C}, 3.47^{\circ}\text{C}]$, indicating that

there is 95% confidence that the true difference in means lies within that interval, and the average air temperature for the lower peninsula is likely a few degrees higher than the average air temperature for the upper peninsula.

The regression for air temperature and population gave a slope estimate of $2.090e-06$, a p-value of $<2e-16$ for this slope estimate, and an R^2 of 0.1026. While the p-value indicates that the slope estimate is statistically significant, the slope estimate is such a small number that it does not show much association between air temperature and population. The R^2 value is also pretty low, and it shows that population only accounts for about 10% of the variation in air temperature. The regression for air temperature and log transformed population gave a slope estimate of 0.8511, a p-value of $<2e-16$ for this slope estimate, and an R^2 of 0.3379. The p-value indicates that the slope estimate is statistically significant, and the slope estimate means that a z-fold increase in population is associated with an average increase in air temperature of $0.85\ln(z)$. The model has also improved to account for around 34% of the variation in air temperature.

Cover Class Regression Results

Predictor	Slope Estimate	P-value for Slope	R^2
Barren Land	-32.78	0.408	0.0105
Log Barren Land	-0.2906	0.222	0.0232
Developed Land	7.8479	$4.35e-08$	0.3717
Log Developed Land	1.8417	$<2e-16$	<u>0.7364</u>
Forests	-6.5577	$1.98e-13$	<u>0.5671</u>
Log Forests	-1.4830	$4.87e-10$	0.4513
Vegetation	5.3450	$2.14e-09$	0.4262
Log Vegetation	1.4018	$1.06e-11$	<u>0.5114</u>
Water	-16.9504	0.0903	0.0435
Log Water	-0.5359	0.0494	0.0581
Wetland	-7.5722	$3.17e-06$	0.2858
Log Wetland	-1.8042	$5.05e-07$	<u>0.3238</u>

For the regression of air temperature and proportion of barren land, neither slope estimate is statistically significant, and neither model accounts for very much variation in air temperature.

For the regression of air temperature and proportion of developed land, both slope estimates are statistically significant and indicate that there is a positive relationship between air

temperature and proportion of developed land. Log transforming the proportion of developed land increased the percent of variation that the model accounts for from 37% to 74%.

For the regression of air temperature and proportion of forests, both slope estimates are statistically significant and indicate that there is a negative relationship between air temperature and proportion of forests. Log transforming the proportion of forests decreased the percent of variation that the model accounts for from 57% to 45%.

For the regression of air temperature and proportion of vegetation, both slope estimates are statistically significant and indicate that there is a positive relationship between air temperature and proportion of vegetation. Log transforming the proportion of vegetation increased the percent of variation that the model accounts for from 43% to 51%.

For the regression of air temperature and proportion of water, the slope estimate for log transformed proportion of water is statistically significant and indicates a negative association between air temperature and proportion of water. The slope estimate for untransformed proportion of water is not statistically significant though, and neither model accounts for very much of the variation in air temperature.

For the regression of air temperature and proportion of wetland, both slope estimates are statistically significant and indicate that there is a negative relationship between air temperature and proportion of wetland. Log transforming the proportion of wetland increased the percent of variation that the model accounts for from 29% to 32%.

The correlation between county air temperature and population is 0.32, with a p-value of $<2.2\text{e-}16$ and 95% confidence interval of [0.25, 0.38]. The p-value indicates that this correlation is statistically significant, and the confidence interval does not include 0, which shows that there is a slight positive relationship between air temperature and population.

Cover Class Correlation Results

Cover Class	Correlation	P-value	95% Confidence Interval
Barren Land	-0.10	0.4084	[-0.33, 0.14]
Developed Land	0.61	4.354e-08	[0.43, 0.74]
Forest	-0.75	1.977e-13	[-0.84, -0.63]
Vegetation	0.65	2.138e-09	[0.49, 0.77]
Water	-0.21	0.0903	[-0.43, 0.03]
Wetland	-0.53	3.166e-06	[-0.69, -0.34]

The correlation between air temperature and proportion of barren land is not statistically significant, and it could be 0. The correlation between air temperature and proportion of developed land is statistically significant and is 0.61, which indicates a positive relationship between these two variables. The correlation between air temperature and proportion of forests is

statistically significant and is -0.75, which indicates a negative relationship between these two variables. The correlation between air temperature and proportion of vegetation is statistically significant and is 0.65, which indicates a positive relationship between these two variables. The correlation between air temperature and proportion of water is not statistically significant, and it could be 0. The correlation between air temperature and proportion of wetland is statistically significant and is -0.53, which indicates a negative relationship between these two variables.

Discussion

These findings show that air temperature in Michigan does not change consistently across the years studied. While the average air temperatures for a few pairs of years were found to be statistically significantly different from the ANOVA and Tukey HSD, many years' average air temperatures were not significantly different from each other, and these results showed no clear progression of air temperature over time.

The Kruskal-Wallis test showed that air temperature does not differ significantly by county, as none of the county medians were found to be significantly different from each other. However, the t-test showed that air temperature does differ significantly between the upper and lower peninsula, with the lower peninsula having a higher temperature on average. This makes sense due to the difference in latitudes.

The regression and correlation for air temperature and population show that population is not a very good predictor of air temperature, and there is not a very strong relationship between these two variables. The slope estimate in the regression for these two variables was very close to 0, and even when population was log transformed, this regression model only accounted for 34% of the variation in air temperature. The correlation between air temperature and population was 0.32, which indicates a positive but not very strong relationship.

The regression and correlation for air temperature and land cover classes show that some land cover classes do have a significant relationship with air temperature. While the barren land and water cover classes do not have significant relationships with air temperature, as seen by the statistically insignificant p-values and low R^2 values in the regression and the statistically insignificant p-values in the correlation, the developed land, forest, vegetation, and wetland cover classes do have a significant relationship with air temperature. The developed land and vegetation cover classes have a positive relationship with air temperature, as seen by the positive slope estimates and positive correlation values. This indicates that an increase in the proportion of developed land or vegetation in a county is associated with an increase in air temperature. The forest and wetland cover classes have a negative relationship with air temperature, as seen by the negative slope estimates and negative correlation values. This indicates that an increase in the proportion of forests or wetlands in a county is associated with a decrease in air temperature.

One limitation of this project is that the monthly average air temperature data used was calculated by averaging the monthly minimum and maximum temperatures. This leaves lots of room for an extreme value to skew this monthly average. An alternative approach would be to

use daily temperature data and take the mean or median of all the days in a month to calculate these monthly values. Another limitation is that this project did not examine adjacent years of data. Using data for years that are five or ten years apart meant that a full time series analysis could not be done, and several years of data were missing from this time frame. More years of air temperature data could be especially useful for the exploration of how air temperature changes over time. Some other considerations are that counties contained varying numbers of stations and stations had varying numbers of monthly observations. Some counties had only one air temperature station, while others had closer to ten. Some stations had the full 120 months of temperature data, while others had just 36. This impacted how much data went into each county calculation, but this project still weighted each county equally. With more careful consideration of this issue, each calculation could be more closely examined to understand exactly what data is being compiled.

Conclusion

While air temperature in Michigan does not seem to vary consistently across years or counties, it is significantly higher in the lower peninsula than the upper peninsula. Although population is not a very strong predictor of air temperature, land cover class can have a stronger relationship with air temperature. The developed land and vegetation cover classes have a positive relationship with air temperature, and the forest and wetland cover classes have a negative relationship with air temperature.

References

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Generative AI Acknowledgement

ChatGPT was used a few times throughout this project. It was first used to match the air temperature stations to which county they are in. ChatGPT was asked, “using this list of station name, latitude, and longitude, please tell me in which county in the state of michigan each station is located.” This was much faster than manually searching up each station or coordinate and cross referencing with state counties. Some of the counties that ChatGPT produced were easy to check just by skimming over, like Alpena Regional Airport being in Alpena and Detroit Metro Airport being in Wayne, and I manually verified a few others by looking up the station name or coordinates and checking against a county map.

ChatGPT was used most heavily for collecting and cleaning the land cover data. It was first asked very generally about “finding land cover data for michigan by county, reading into R.” After looking at the database it suggested and choosing and downloading a file, its instructions were followed for extracting the data for Michigan counties, converting it to a table, and looking up the cover classes.

ChatGPT was also used for help with creating maps. For plotting the station locations on the map of Michigan, it was asked, “using a dataset of latitude and longitude coordinates, how to create a map of michigan with county lines and the coordinates plotted on the map. already have a dataset outlining michigan counties.” For creating the map of counties by land cover category, it was asked, “now how to make map of michigan where each county is shaded depending on a categorical variable. have a dataset of county name and category.” The other shaded maps were created using similar methods to the land cover category map.

ChatGPT was used for some help with smaller data cleaning and compilation tasks. For converting the temperature values from Celsius to Fahrenheit, it was asked, “in data formatted like this, how to convert columns emnt, emxt, tavg, tmin, tmax from fahrenheit to celcius for year 1970 and months 1-6.” Its conversion formula and the output produced were double checked. For joining the population data with the yearly county temperature data, it was asked, “how to add a column of population values that gives the population for that county and year, from a dataset formatted like this.” Its output was double checked and tweaked accordingly. Some of the processes it suggested for cleaning and joining data were then modified to be used in other situations.